

# Adding and Subtracting in Scientific Notation

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**MA.8.NSO.1.5**

*I can add and subtract two numbers written in scientific notation, and write the answer in scientific notation.*

*Last unit you learned to write numbers in scientific notation.*

**Today you put two of them together — and the powers of ten have to match first.**

# Warm-Up

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*On your own. Four minutes. No notes.*

1. Write 0.00062 in scientific notation.
2. How many times larger is  $8 \times 10^7$  than  $2 \times 10^4$  ?
3. Find the value:  $\sqrt{144}$
4. Find the sum:  $0.49 + 1.2$

# Warm-Up

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*Answers.*

1. Write 0.00062 in scientific notation.  **$6.2 \times 10^{-4}$**
2. How many times larger is  $8 \times 10^7$  than  $2 \times 10^4$  ?  
**4,000**
3. Find the value:  $\sqrt{144}$  **12**
4. Find the sum:  $0.49 + 1.2$  **1.69**

# Notes

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*Copy the table.*

## I. A power of ten is a place value

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| $10^3$    | $10^4$        | $10^5$            | $10^6$   |
|-----------|---------------|-------------------|----------|
| thousands | ten thousands | hundred thousands | millions |

$$49,500 + 121,000 = 170,500$$

$$4.95 \times 10^4 + 12.1 \times 10^4 = 17.05 \times 10^4 = 1.705 \times 10^5$$

**Two numbers can be added only when they carry the same power of ten.**

# Notes

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## II. Same power: add or subtract the coefficients

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$$(5.2 \times 10^6) + (3.1 \times 10^6) = 8.3 \times 10^6$$

$$(7 \times 10^{-8}) + (6 \times 10^{-8}) = 13 \times 10^{-8} = 1.3 \times 10^{-7}$$

**Add or subtract the coefficients. The power of ten does not change.**

Then check the coefficient: if it is not between 1 and 10, the answer is not finished.

# Notes

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## III. Different powers: rewrite one of them first

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$$(9.4 \times 10^5) - (8.2 \times 10^3)$$

$$= (9.4 \times 10^5) - (0.082 \times 10^5) = 9.318 \times 10^5$$

$$(1.3 \times 10^3) + (3.4 \times 10^5) = 3.413 \times 10^5 \quad , \quad \text{never} \quad 4.7 \times 10^8$$

**Rewrite one number so both carry the same power of ten, then add or subtract the coefficients.**

Exponent up one, decimal point one place left. The value does not move.

# Example 1

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*Same power, and the answer is not finished.*

$$(8.6 \times 10^7) + (5.9 \times 10^7)$$

**Add. Write the answer in scientific notation.**

# Example 1

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*Worked.*

$$(8.6 \times 10^7) + (5.9 \times 10^7) = 14.5 \times 10^7 \quad \text{same power: add the coefficients}$$

$$14.5 \times 10^7 = 1.45 \times 10^8 \quad \text{14.5 is not less than 10: point one place left, exponent up one}$$

$$145,000,000 \quad \text{check in standard form}$$

**Answer:  $1.45 \times 10^8$**

# Your Turn

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*Same steps, your numbers. Boards up when done.*

$$(4.8 \times 10^{-3}) + (7.5 \times 10^{-3})$$

# Your Turn

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*Answer.*

$$(4.8 \times 10^{-3}) + (7.5 \times 10^{-3})$$

$12.3 \times 10^{-3}$ , then repair

**Answer:  $1.23 \times 10^{-2}$**

# Example 2

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*Different powers, and a subtraction.*

$$(7.3 \times 10^6) - (4.5 \times 10^5)$$

**Subtract. Write the answer in scientific notation.**

# Example 2

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*Worked.*

$$4.5 \times 10^5 = 0.45 \times 10^6 \quad \text{rewrite over } 10^6$$

$$(7.3 \times 10^6) - (0.45 \times 10^6) = 6.85 \times 10^6 \quad \text{same power: subtract the coefficients}$$

$$7,300,000 - 450,000 = 6,850,000 \quad \text{check in standard form}$$

**Answer:  $6.85 \times 10^6$**

# Your Turn

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*Same steps, your numbers. Boards up when done.*

$$(5.6 \times 10^{-4}) - (9.0 \times 10^{-6})$$

# Your Turn

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*Answer.*

$$(5.6 \times 10^{-4}) - (9.0 \times 10^{-6})$$

$0.09 \times 10^{-4}$  comes off  $5.6 \times 10^{-4}$

**Answer:  $5.51 \times 10^{-4}$**

# Whiteboards • Question 1 of 9

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*Boards up on three.*

$$(3.2 \times 10^5) + (4.1 \times 10^5)$$

**Answer it.**

*Same power.*

# Whiteboards • Question 1 of 9

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*Answer.*

$$(3.2 \times 10^5) + (4.1 \times 10^5)$$

add the coefficients

**Answer:  $7.3 \times 10^5$**

# Whiteboards • Question 2 of 9

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*Boards up on three.*

A stadium sold  $4.2 \times 10^4$  tickets on Friday  
and  $3.5 \times 10^4$  on Saturday. Each ticket cost 38 dollars.

How many tickets were sold on the two days?

**Answer it.**

*Same power.*

# Whiteboards • Question 2 of 9

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*Answer.*

A stadium sold  $4.2 \times 10^4$  tickets on Friday  
and  $3.5 \times 10^4$  on Saturday. Each ticket cost 38 dollars.

How many tickets were sold on the two days?

$$4.2 + 3.5$$

**Answer:  $7.7 \times 10^4$**

# Whiteboards • Question 3 of 9

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*Boards up on three.*

$$(6 \times 10^{-5}) + (7 \times 10^{-5})$$

**Answer it.**

*Finish the answer.*

# Whiteboards • Question 3 of 9

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*Answer.*

$$(6 \times 10^{-5}) + (7 \times 10^{-5})$$

$13 \times 10^{-5}$  is not finished

**Answer:  $1.3 \times 10^{-4}$**

# Whiteboards • Question 4 of 9

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*Boards up on three.*

$$(5.6 \times 10^8) - (4.9 \times 10^8)$$

**Answer it.**

*Finish the answer.*

# Whiteboards • Question 4 of 9

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*Answer.*

$$(5.6 \times 10^8) - (4.9 \times 10^8)$$

$0.7 \times 10^8$  is not finished either

**Answer:  $7 \times 10^7$**

# Whiteboards • Question 5 of 9

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*Boards up on three.*

$$(1.3 \times 10^3) + (3.4 \times 10^5)$$

Which is the sum?

**A.**  $3.413 \times 10^5$

**B.**  $4.7 \times 10^8$

**C.**  $4.7 \times 10^5$

**D.**  $3.53 \times 10^5$

**Answer it.**

# Whiteboards • Question 5 of 9

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*Answer.*

$$(1.3 \times 10^3) + (3.4 \times 10^5)$$

Which is the sum?

**A.  $3.413 \times 10^5$**

**B.  $4.7 \times 10^8$**

**C.  $4.7 \times 10^5$**

**D.  $3.53 \times 10^5$**

**Answer: A —  $3.413 \times 10^5$**

# Whiteboards • Question 6 of 9

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*Boards up on three.*

$$(2.5 \times 10^6) + (9.8 \times 10^6)$$

Which is the sum, written in scientific notation?

**A.**  $1.23 \times 10^7$

**B.**  $12.3 \times 10^6$

**C.**  $12.3 \times 10^{12}$

**D.**  $1.23 \times 10^6$

**Answer it.**

# Whiteboards • Question 6 of 9

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*Answer.*

$$(2.5 \times 10^6) + (9.8 \times 10^6)$$

Which is the sum, written in scientific notation?

**A.  $1.23 \times 10^7$**

**B.  $12.3 \times 10^6$**

**C.  $12.3 \times 10^{12}$**

**D.  $1.23 \times 10^6$**

**Answer: A —  $1.23 \times 10^7$**

# Whiteboards • Question 7 of 9

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*Boards up on three.*

$$(7.8 \times 10^{-2}) - (6.4 \times 10^{-3})$$

**Answer it.**

*Rewrite one of them first.*

# Whiteboards • Question 7 of 9

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*Answer.*

$$(7.8 \times 10^{-2}) - (6.4 \times 10^{-3})$$

$$6.4 \times 10^{-3} = 0.64 \times 10^{-2}$$

**Answer:  $7.16 \times 10^{-2}$**

# Whiteboards • Question 8 of 9

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*Boards up on three.*

$$(4.5 \times 10^6) + (n \times 10^6) = 1.02 \times 10^7$$

What is the value of  $n$  ?

**Answer it.**

*Rewrite the answer over  $10^6$  first.*

# Whiteboards • Question 8 of 9

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*Answer.*

$$(4.5 \times 10^6) + (n \times 10^6) = 1.02 \times 10^7$$

What is the value of  $n$  ?

$$1.02 \times 10^7 = 10.2 \times 10^6$$

**Answer:  $n = 5.7$**

# Whiteboards • Question 9 of 9

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*Take your time. Boards up when you have written it.*

Devon writes  $(1.3 \times 10^3) + (3.4 \times 10^5) = 4.7 \times 10^8$  .

Name the two mistakes he made, and give the correct sum.

**Write your answer in sentences.**

*Two mistakes, then the sum.*

# Whiteboards • Question 9 of 9

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*Answer.*

Devon writes  $(1.3 \times 10^3) + (3.4 \times 10^5) = 4.7 \times 10^8$ .

Name the two mistakes he made, and give the correct sum.

one mistake in the coefficients, one in the exponents

**Answer:  $3.413 \times 10^5$**

# Independent Set

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*Six questions. On your own, in silence.*

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

# IXL

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*Last five minutes.*

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

## **Today's skills — all required**

Add and subtract numbers written in scientific notation — HUR